Tutorial No.: 8
Engineering Mechanics (ME-101)
Date: 18/03/2024 and Time: 7.55 to 8.55 AM

Question 1: The 10-kg steel sphere is suspended from the 15-kg frame which slides down the 20^0 incline, as shown in Figure 1. If the coefficient of kinetic friction between the frame and incline is 0.15, compute the tension in each of the supporting wires A and B. Take $g = 9.81$ m/s^2. **[10 Marks]**

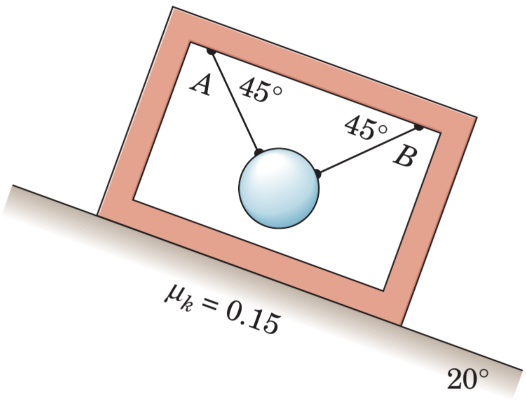
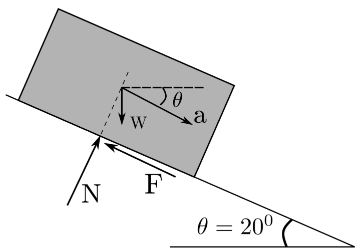
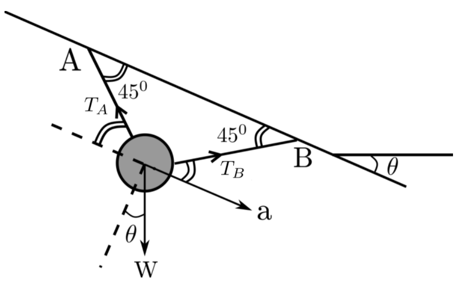


Figure 1

Solution 1:



[1 mark]



[1 mark]

Considering the sphere as well as the frame as a system one can write,

$$mg\sin\theta - F = ma$$

$$F = \mu_k N = \mu_K mg\cos\theta$$

$$\Rightarrow a = g(\sin\theta - \mu_k\cos\theta) = 1.972 \text{ m/s}^2$$ **[2 marks]**

Writing the equation of motion parallel and normal the slope, respectively, one gets,

$$T_B\cos\frac{\pi}{4} - T_A\cos\frac{\pi}{4} + w\sin\theta = \frac{w}{g}a$$ **[2 marks]**

$$T_B\sin\frac{\pi}{4} + T_A\sin\frac{\pi}{4} - w\cos\theta = 0$$ **[2 marks]**

$$\Rightarrow T_B - T_A = \sqrt{2}(ma - mg\sin 20^0)$$

$$T_B - T_A = -19.561$$

$$\Rightarrow T_B + T_A = \sqrt{2}mg\cos\theta$$

$$T_B + T_A = 130.367$$

Thus, $\quad T_A = 55.403$ N $\quad$ and $\quad T_B = 74.964$ N **[2 marks]**

Question 2: The 2-kg slider fits loosely in the smooth slot of the disk, which rotates about a vertical axis through point O, as illustrated in Figure 2. The slider is free to move slightly along the slot before one of the wires becomes taut. If the disk starts from rest at time $t = 0$ and has a constant clockwise angular acceleration of 0.5 rad/s^2, plot the tensions in wires 1 and 2 and the magnitude N of the force normal to the slot as functions of time t for the interval $0 \leq t \leq 5$ s. **[10 Marks]**

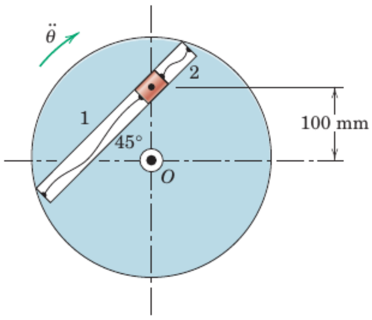
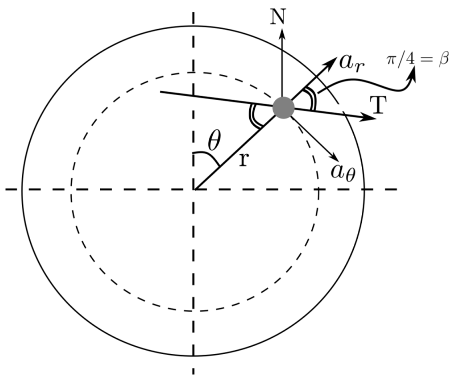


Figure 2

Solution 2: $\alpha = 0.5$ rad/s^2 (const.), θ =1/2 αt^2, the slot is smooth and slider can move slightly along the slot. Assuming tension in wire 2 as positive initially,



[2 marks]

Invoking the equation of motion along the slot and perpendicular to the slot, respectively,

$$T = m(a_r \cos\beta + a_\theta \sin\beta)$$

$$N = m(a_r \sin\beta - a_\theta \cos\beta)$$

At $\beta = \pi/4$, $a_r = \cancelto{0}{\ddot{r}} - r\dot{\theta}^2 = -r\dot{\theta}^2 = -r\alpha^2 t^2$

$$a_\theta = r\ddot{\theta} + 2\cancelto{0}{\dot{r}}\dot{\theta} = r\alpha$$ **[2 marks]**

$$T = mr(-\alpha^2 t^2 + \alpha)\frac{1}{\sqrt{2}} = \frac{mr\alpha}{\sqrt{2}}(1 - \alpha t^2)$$

$$N = mr(-\alpha^2 t^2 - \alpha)\frac{1}{\sqrt{2}} = -\frac{mr\alpha}{\sqrt{2}}(1 + \alpha t^2)$$

[2 marks]

Tension in the wire 2 becomes zero as,

$$T = 0 \quad \Rightarrow \alpha t^2 = 1 \quad \Rightarrow t^2 = 1/\alpha \quad \Rightarrow t^2 = 1/0.5 \quad \Rightarrow t = \sqrt{2}$$

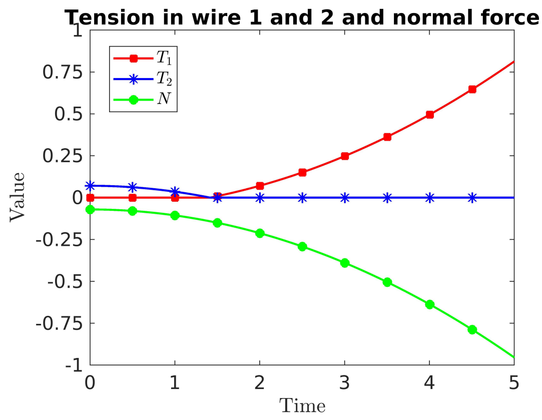
Thus, for $t \leqslant \sqrt{2}$, $\quad T_2 = \dfrac{mr\alpha}{\sqrt{2}}(1 - \alpha t^2)$ and $T_1 = 0$

$$N = -\frac{mr\alpha}{\sqrt{2}}(1 + \alpha t^2)$$ **[1 mark]**

for $t > \sqrt{2}$, $\quad T_1 = \dfrac{mr\alpha}{\sqrt{2}}(\alpha t^2 - 1)$ and $T_2 = 0$

$$N = -\frac{mr\alpha}{\sqrt{2}}(1 + \alpha t^2)$$ (Here, N remains same irrespective of time t)

[1 mark]



[2 marks]

Question 3: The spring attached to the 10-kg mass is non-linear, having the force–deflection relationship shown in the Figure 3, and is unstretched when $x = 0$. If the mass is moved to the position $x = 100$ mm and released from rest, determine its velocity v when $x = 0$. Determine the corresponding velocity v' if the spring would have been linear according to $F = 4x$, where x is in meter and the force F is in kN. **[10 Marks]**

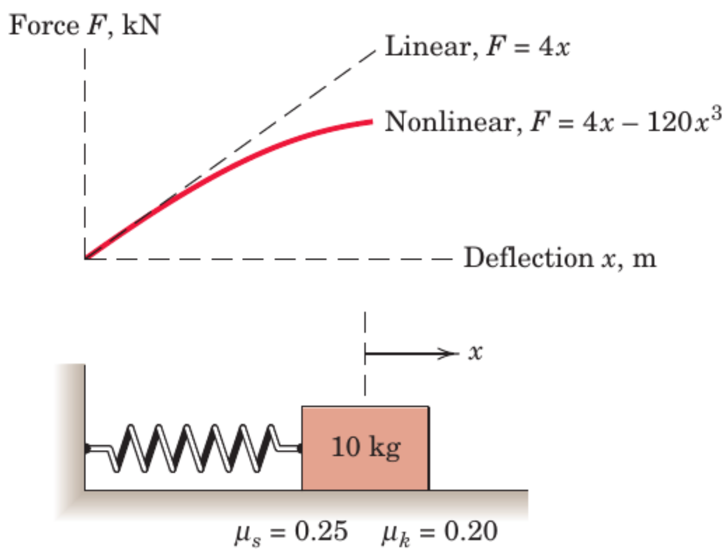
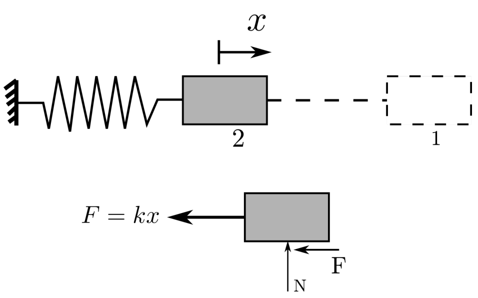


Figure 3

Solution 3:



$$T_1 + V_1 + U_{1-2} = T_2 + V_2$$

$$\implies 0 + V_1 - \mu_k mg(x_1 - x_2) = \frac{1}{2}m{v^2}_2 + 0$$ **[3 marks]**

$$V_1 = \int_2^1 F dx$$

$$= \int_0^x (4x - 120x^3)dx = (2x^2 - 30x^4) \times 10^3 \text{ J}$$ **[3 marks]**

Thus, $\frac{1}{2} \times 10 \times {v^2}_2 = (2{x^2}_1 - 30{x^4}_1) \times 10^3 - 0.2 \times 10 \times 9.81 \times x_1$

In non-linear case,

$$\implies {v_2}^2 = \frac{1}{5}[2000 \times 0.1^2 - 30 \times 0.1^4 \times 1000 - 2 \times 9.81 \times 0.1]$$

$$\implies v_2 = 1.734 \text{ m/s}$$ **[2 marks]**

In linear case,

$${v_2}^2 = \frac{1}{5}[2000 \times 0.1^2 - 2 \times 9.81 \times 0.1]$$

$$\implies v_2 = 1.899 \text{ m/s}$$ **[2 marks]**

Question 4: A small mass particle is given an initial velocity v_0 tangent to the horizontal rim of a smooth hemispherical bowl at a radius r_0 from the vertical centerline, as shown at point A in Figure 4. As the particle slides past point B, a distance h below A and a distance r from the vertical centerline, its velocity v makes an angle θ with the horizontal tangent to the bowl through B. Determine θ. **[10 Marks]**

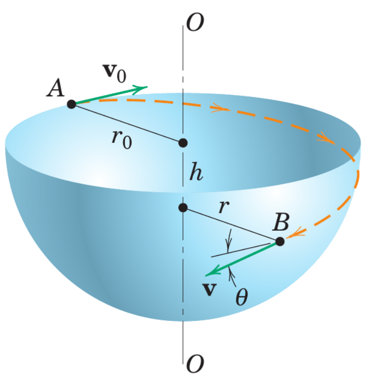


Figure 4

Solution 4:

$$[(H_O)_1 = (H_O)_2 \implies mv_0r_0 = mvr\cos\theta$$ **[4 marks]**

Also, energy is conserved so that $E_1 = E_2$.Thus

$$[T_1 + V_1 = T_2 + V_2] \implies \frac{1}{2}mv_0^2 + mgh = \frac{1}{2}mv^2 + 0$$ **[4 marks]**

$$v = \sqrt{v_0^2 + 2gh}$$

Eliminating v and substituting $r^2 = r_0^2 - h^2$gives

$$v_0r_0 = \sqrt{v_0^2 + 2gh}\sqrt{r_0^2 - h^2}\cos\theta$$

$$\theta = \cos^{-1}\frac{1}{\sqrt{1+\frac{2gh}{v_0^2}}\sqrt{1-\frac{h^2}{r_0^2}}}$$ **[2 marks]**

Question 5: Three steel spheres of equal weight are suspended from the ceiling by cords of equal length which are spaced at a distance slightly greater than the diameter of the spheres (refer Figure 5). After being pulled back and released, sphere A hits sphere B, which then hits sphere C. Say, the coefficient of restitution between the spheres is denoted by e, and the velocity of A just before it hits B is v_0, determine, (a) the velocities of A and B immediately after the first collision, (b) the velocities of B and C immediately after the second collision. (c) Assuming that the n spheres are suspended from the ceiling and that the first sphere is pulled back and released as described above, determine the velocity of the last sphere after it is hit for the first time. (d) Use the result of part c to obtain the velocity of the last sphere when n = 5 and e = 0.9. **[10 Marks]**

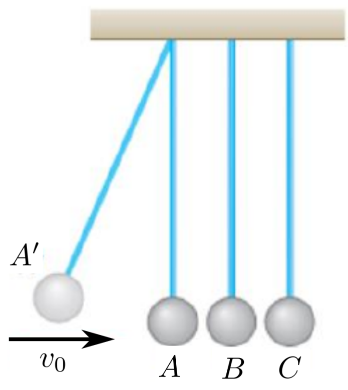
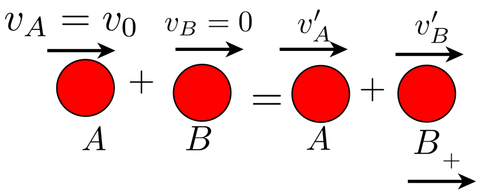


Figure 5

Solution 5: (a) First collision (**between A and B**): The total momentum is conserved:



$$
\begin{aligned}
mv_A + mv_B &= mv'_A + mv'_B \\
v_0 &= v'_A + v'_B
\end{aligned}
\tag{1}
$$

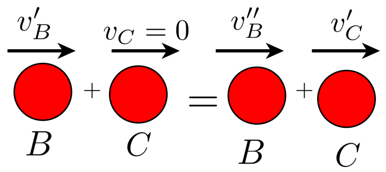
Relative velocities:

$$
\begin{aligned}
(v_A - v_B)e &= (v'_B - v'_A) \\
v_0 e &= v'_B - v'_A
\end{aligned}
\tag{2}
$$

Solving equations (1) and (2) simultaneously,

$$v'_A = \frac{v_0(1-e)}{2} \quad \text{and} \quad v'_B = \frac{v_0(1+e)}{2}$$ **[3 marks]**

(b) Second collision (**between B and C**): The total momentum is conserved.



$$mv'_B + mv_C = mv''_B + mv'_C$$

Using the result from (a) for v'_B

$$\frac{v_0(1+e)}{2} + 0 = v''_B + v'_C \tag{3}$$

Relative velocities:

$$(v'_B - 0)e = v'_C - v''_B$$

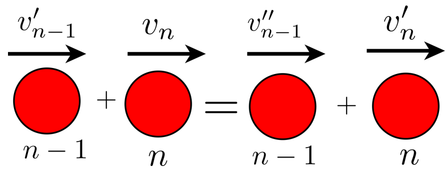
Substituting again for v'_B from (a)

$$\frac{v_0(1+e)e}{2} = v'_C - v''_B \tag{4}$$

Solving equations (3) and (4),

$$
\begin{aligned}
v'_C &= \frac{v_0(1+e)^2}{4} \\
v''_B &= \frac{v_0(1-e^2)}{4}
\end{aligned}
$$ **[3 marks]**

(c) Observing the velocity expressions of v'_B and v'_C, after first and second collisions, respectively, one can formulate the velocity of n^{th} sphere after $(n-1)$ collisions.



$$v'_n = \frac{v_0(1+e)^{(n-1)}}{2^{(n-1)}}$$ **[2 marks]**

(d) For n=5, e=0.90, from the answer part to part (c) with $n = 5$

$$v'_5 = \frac{v_0(1+0.9)^{(5-1)}}{2^{(5-1)}}$$

$$\implies v'_5 = \frac{v_0(1.9)^4}{2^4} = 0.815v_0$$ **[2 marks]**

Question 6: As shown in Figure 6, a 10 kg package drops from a chute into a 20 kg cart with a velocity of 3 m/s. Knowing that the cart is initially at rest and can roll freely, determine, (a) the final velocity of the cart, (b) the impulse exerted by the cart on the package, (c) the fraction of the initial energy lost in the impact. Assume that there is no friction between cart and floor, and the distance from chute to wall of the cart is short, so that the package hits the cart at 3m/s and it is not like a projectile motion. **[10 Marks]**

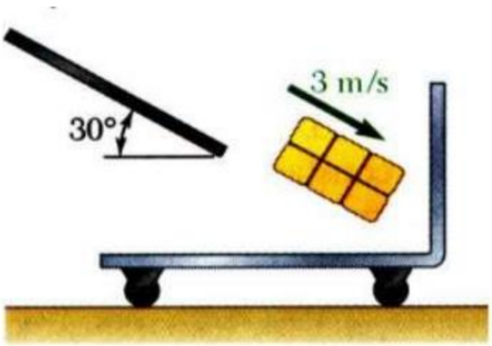
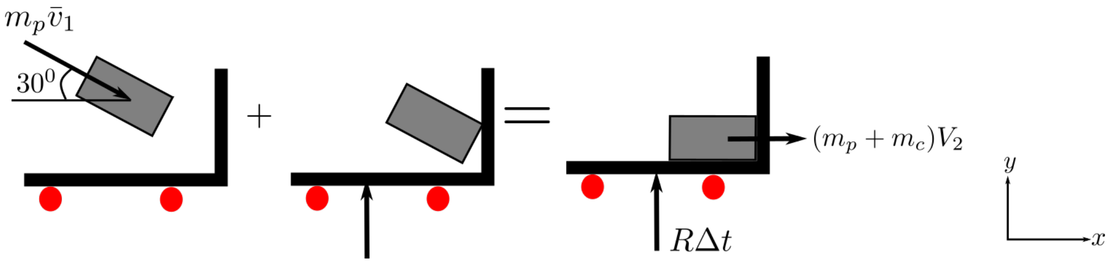


Figure 6

Solution 6 : (a) Applying the impulse-momentum principle,



$$m\vec{v}_1 + I_{1\to 2} = (m_p + m_c)\vec{v}_2$$

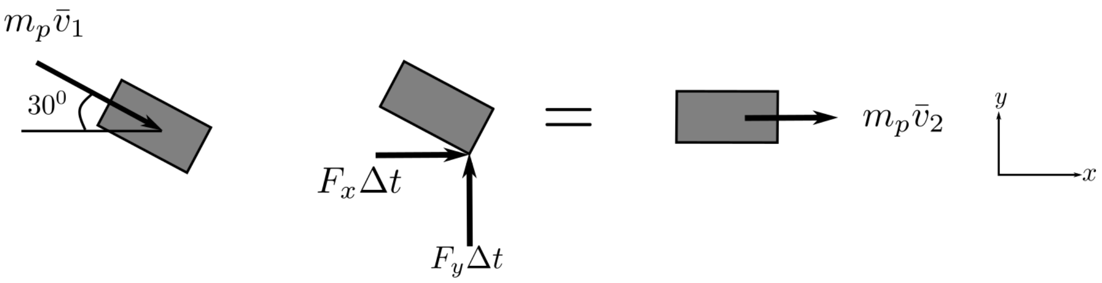
$$\implies 10 \times (3\cos 30^0\hat{i} - 3\sin 30^0\hat{j}) + R\delta t\hat{j} = (10+20)v_2\hat{i}$$

$$25.98\hat{i} - 15\hat{j} + R\delta t\hat{j} = 30v_2\hat{i}$$

$$\text{Equating } \hat{i} \text{ component on obtains, } v_2 = \frac{25.98}{30} = 0.866 \text{ m/s}$$

[3 marks]

(b) Applying the impulse-momentum equation to the package only,



$$m_p\vec{v}_1 + I_{1\to 2} = m_p\vec{v}_2$$

$$\implies 10 \times (3\cos 30^0\hat{i} - 3\sin 30^0\hat{j}) + (F_x\hat{i} + F_y\hat{j})\delta t = 10v_2\hat{j}$$

$$25.98\hat{i} - 15\hat{j} + (F_x\hat{i} + F_y\hat{j})\delta t = 10 \times 0.866\hat{i}$$

[3 marks]

Equating terms in $\hat{i}$ and $\hat{j}$ components separately one obtains

$$F_x\delta t = 8.66 - 25.98 = -17.32 \text{ Ns}$$

$$F_y\delta t = 15 \text{ Ns}$$

$$\text{Impulse} = F\delta t = \sqrt{(-17.32)^2 + 15^2} = 22.91 \text{ Ns}$$

[2 marks]

(c) To find the fraction of energy loss,

$$T_1 = \frac{1}{2}m_p v_1^2 = \frac{1}{2} \times 10 \times 3^2 = 45 \text{ J}$$

$$T_2 = \frac{1}{2}(m_p + m_c)v_2^2 = \frac{1}{2} \times 30 \times 0.866^2 = 11.25 \text{ J}$$

$$\% \text{ of energy loss} = \frac{T_1 - T_2}{T_1} \times 100 = \frac{45 - 11.25}{45} \times 100 = 75.$$

[2 marks]